

Project Proposal:

Adaptive Sample Allocation in Multilevel Monte Carlo for Option Pricing

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1 Project Scope

Our project studies how Multilevel Monte Carlo (MLMC) can be used to improve the computational efficiency of option pricing. The main motivation is that standard Monte Carlo (MC) simulation is flexible and easy to apply, but it can become computationally expensive when high accuracy is required, especially for path-dependent options. We aim to begin from the classical Black–Scholes setting, reproduce baseline results using standard Monte Carlo, and then move to MLMC for both European and path-dependent options.

The project will proceed in five stages. First, we will study the Black–Scholes model and standard Monte Carlo pricing, and reproduce benchmark results for European options. Second, we will learn and implement the MLMC algorithm for European option pricing. Third, we will extend the MLMC framework to a path-dependent option, most likely an arithmetic Asian option. Fourth, we will propose a modification to the original MLMC sample allocation rule by introducing an adaptive allocation strategy based on iterative variance estimation across levels. Finally, we will compare the modified MLMC algorithm with the original MLMC in terms of pricing accuracy, variance, root mean squared error, and computational cost.

Our project is designed to match the three required elements of the course project. The *learning* component is the study of Black–Scholes pricing, standard Monte Carlo simulation, and the MLMC algorithm. The *analysis* component is the empirical comparison of standard MC, original MLMC, and modified MLMC across different option types and parameter settings. The *research* component is the modification of the MLMC sample allocation rule and the evaluation of whether this modification improves efficiency in practice.

2 Background and Motivation

Under the Black–Scholes framework, the underlying asset price follows geometric Brownian motion. For a European call option, the arbitrage-free price has a closed-form expression, which makes it an ideal benchmark for validating simulation-based pricing methods. In standard Monte Carlo pricing, the option value is estimated by simulating many asset paths and averaging discounted payoffs. However, for a target root mean squared error of order ε , standard Monte Carlo with Euler discretization can have computational cost of order $O(\varepsilon^{-3})$.

MLMC reduces this cost by combining estimators across multiple discretization levels. Instead of simulating only on the finest grid, MLMC writes the finest-level expectation as a

telescoping sum:

$$\mathbb{E}[\hat{P}_L] = \mathbb{E}[\hat{P}_0] + \sum_{\ell=1}^L \mathbb{E}[\hat{P}_\ell - \hat{P}_{\ell-1}].$$

The idea is that coarse levels are cheap and capture most of the expectation, while fine levels estimate small corrections with relatively low variance if the fine and coarse paths are properly coupled. This makes MLMC especially attractive for option pricing problems in which standard Monte Carlo becomes too costly.

3 Methods

3.1 Stage 1: Black–Scholes and Standard Monte Carlo

We will first implement the Black–Scholes formula for European call options under geometric Brownian motion and compare it against standard Monte Carlo simulation. This will provide:

- a benchmark for validating simulation code,
- a reference point for pricing accuracy,
- and a baseline for later computational comparisons.

The asset price dynamics under the risk-neutral measure are

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

For a European call option with strike K and maturity T , the price is

$$V_0 = e^{-rT} \mathbb{E}[(S_T - K)^+].$$

3.2 Stage 2: MLMC for European Options

Next, we will implement MLMC for European option pricing. We will construct coupled fine and coarse paths using the same Brownian path, estimate correction terms $\hat{P}_\ell - \hat{P}_{\ell-1}$, and allocate samples across levels according to the standard MLMC sample allocation rule. We will compare standard MC and MLMC in terms of:

- estimated option values,
- variance,
- root mean squared error (RMSE),
- and computational cost.

3.3 Stage 3: Extension to a Path-Dependent Option

We will then extend the MLMC framework to a path-dependent option, most likely an arithmetic Asian option. This is a natural next step because MLMC becomes more meaningful when no simple closed-form benchmark exists. For the Asian option, the payoff depends on the average price along the path rather than only the terminal price, making the simulation problem more challenging and more representative of realistic applications.

3.4 Stage 4: Modification of the Sample Allocation Rule

The main research contribution of our project will be a modification of the MLMC sample allocation strategy. In the original algorithm, the number of samples on each level depends on estimated level variances and costs. Our proposed modification is to make this allocation more adaptive by updating the sample sizes iteratively as improved variance estimates become available. Intuitively, this may reduce unnecessary sampling on levels whose correction variance is already stable or negligible.

At a high level, our modified approach will:

1. start with a pilot sample on each level,
2. estimate level variances,
3. allocate additional samples according to current variance estimates,
4. update the variance estimates again,
5. and repeat until the target accuracy is reached.

3.5 Stage 5: Comparison and Evaluation

Finally, we will compare:

- standard Monte Carlo,
- original MLMC,
- and modified MLMC.

The main evaluation metrics will include:

- pricing error,
- standard error,
- RMSE,
- runtime,
- total computational cost,
- and level-by-level sample allocation.

Our key research question is:

Can an adaptive sample allocation strategy improve the practical efficiency of MLMC for option pricing compared with the original allocation rule?

4 Expected Outcomes

We expect the following outcomes:

- Standard MLMC should outperform standard Monte Carlo in computational efficiency for a fixed target accuracy.
- The performance gain should be more visible for path-dependent options than for simple European options.
- Our adaptive allocation modification may reduce computational cost further by distributing sampling effort more effectively across levels.

We also recognize that the modified allocation may not always outperform the original method. If so, that would still be a meaningful result, since part of the project is to test whether a plausible algorithmic change actually improves performance.

5 Contribution Matrix

Member	Responsibility
Yongxu Wang	Black–Scholes formula implementation, standard Monte Carlo benchmark, write-up of background and motivation
Yinan Shi	MLMC implementation for European option, coupled path construction, initial experiments
Alex Zhu	Path-dependent option implementation, adaptive sample allocation modification, comparison figures and tables
All Members	Discussion of results, slide preparation, final report revision, and presentation rehearsal

If our group has a different number of members, we will adjust this matrix accordingly while keeping responsibilities concrete and clearly separated.

6 Logistics

We propose the following meeting schedule:

- **Meeting 1 4/23:** finalize scope, references, and task assignment
- **Meeting 2 4/27:** review Black–Scholes and standard MC implementation
- **Meeting 3 4/30:** integrate MLMC implementation and test European option results
- **Meeting 4 5/5:** implement and evaluate path-dependent option and modified allocation
- **Meeting 5 5/8:** finalize report, figures, and presentation slides
- **Meeting 6 5/10:** presentation rehearsal

We plan to use GitHub/Colab for version control, Overleaf for collaborative writing, and shared online documents for notes and meeting records.

7 Initial References

1. Michael B. Giles, *Multilevel Monte Carlo path simulation*, Oxford-Man Institute Working Paper, 2007.
2. Desmond J. Higham, *An introduction to multilevel Monte Carlo for option valuation*, International Journal of Computer Mathematics, 92(12):2347–2360, 2015.